d3507a8c-732b-4ab6-9220-18c8f569ca

<current>

<request>

You are working on an issue in the 'musicmonk42/VulcanAMI_LLM' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/VulcanAMI_LLM/VulcanAMI_LLM (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-01-18T00:02:48.433Z</current_datetime>

Consider the following problem statement:

<problem_statement>
Problem

When VULCAN's reasoning engines return structured conclusions (like SAT solver results), OpenAI responds with useless meta-commentary instead of presenting the actual answer.

Current behavior:
- Symbolic SAT solver correctly returns: `{"satisfiable": false, "result": "NO", "proof": "1. From ¬C: C = False\n2. From B→C..."}`
- This gets sent to OpenAI for formatting
- OpenAI responds: "VULCAN successfully processed your query with confidence 0.85"

Expected behavior:
- OpenAI should respond: "NO. The set is unsatisfiable. Proof: 1. From ¬C: C = False..."

Root Cause

In `src/vulcan/llm/hybrid_executor.py`, the user prompt sent to OpenAI (around line 3080) says:

```python
```

````
user_prompt = f"""Format this VULCAN reasoning output for the
user.

VULCAN's reasoning output (this is what VULCAN computed):
{output_json}

Write a natural, helpful response that explains VULCAN's results
and conclusions.
Do NOT add any analysis or reasoning beyond what is in VULCAN's
output."""
```

This prompt doesn't explicitly tell OpenAI to:
1. Find the `conclusion` field in the JSON
2. Present its contents as the actual answer
3. Not just acknowledge that processing happened

## Fix Required

Update the user prompt in `_format_vulcan_output_with_openai`
method to explicitly instruct OpenAI to:

1. **Extract the `conclusion` field** from the nested JSON
structure (it may be at `result.agent_reasoning.conclusion` or
`result.unified.conclusion` or similar paths)
2. **Present the conclusion as the primary answer** - if it
contains `result: "NO"` or `satisfiable: false`, say "NO"
clearly
3. **Include supporting details** like `proof`, `explanation`,
`reasoning_steps` if present
4. **Never respond with just meta-commentary** like "VULCAN
processed with confidence X" - always present the actual answer

Example prompt fix:

```python
user_prompt = f"""You are formatting VULCAN's reasoning output
into a clear answer for the user.

VULCAN's structured output:
{output_json}

CRITICAL INSTRUCTIONS:
1. Find the 'conclusion' field in the JSON (may be nested under
'result', 'agent_reasoning', 'unified', etc.)
2. The conclusion IS the answer - present it clearly and
directly
````

````
3. If the conclusion contains 'result', 'answer', 'satisfiable',
or similar fields, state that value explicitly
4. If there's a 'proof', 'explanation', or 'reasoning_steps',
include them to support the answer
5. NEVER respond with just "VULCAN processed with confidence X"
- that's not an answer
6. Start your response with the actual answer/conclusion, then
provide supporting details

Format the answer naturally but ensure the actual conclusion is
prominently stated."""
```

## Files to Modify

- `src/vulcan/llm/hybrid_executor.py` - Update the user prompt
in `_format_vulcan_output_with_openai` method (around line 3080)

## Test Cases

After the fix, these queries should return actual answers:

1. SAT query "Is A→B, B→C, ¬C, AvB satisfiable?" → Should return
"NO. The set is unsatisfiable. Proof: ..."
2. Mathematical query → Should return the computed result, not
"VULCAN computed with confidence 0.9"
3. Probabilistic query → Should return "P(X|+) = 0.167" not
"Analysis complete with confidence 0.95"

</problem_statement>
````

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.
````